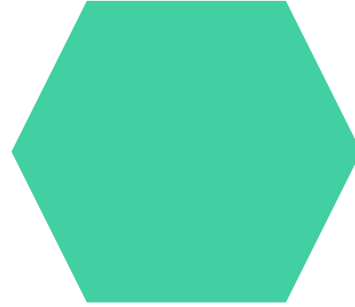




Project Title – Seasonal Agriculture Performance Analysis

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PROBLEM STATEMENT

- Agricultural performance varies significantly across seasons due to changes in **environmental conditions, farming practices, resource availability, and market conditions**. However, raw agricultural data does not clearly explain how these factors influence crop yield, production, resource efficiency, and profitability across different seasons.
- The problem is to analyze the agricultural dataset and identify meaningful seasonal patterns, trends, relationships, and variations in agricultural performance, so that differences between seasons can be understood and used to support better agricultural planning and decision-making.



WHO ARE THE END USERS?

- **Farmers** - to understand seasonal crop performance, yield, resource usage, and profitability.
- **Agricultural Planners & Advisors** - to support seasonal crop planning and resource allocation.
- **Government & Agriculture Departments** - to support evidence-based agricultural policies and programs.
- **Agribusiness & Input Suppliers** - to understand seasonal demand for irrigation, fertilizers, pesticides, and other inputs.
- **Researchers & Data Analysts** - to study relationships between environmental conditions, farming practices, and agricultural outcomes.

In short: The project helps stakeholders make **data-driven decisions** for **better seasonal agricultural planning, resource efficiency, and profitability.**

Technology Used

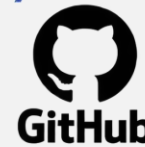
Programming & Analysis

- Python
- Pandas
- NumPy
- SciPy



Data Visualization

- Matplotlib
- Seaborn



Development Environment

- **Google Colab** - for running and documenting the complete Jupyter Notebook analysis.

Documentation & Version Control

- **GitHub** - for storing and sharing the project notebook, dataset, code, and documentation.

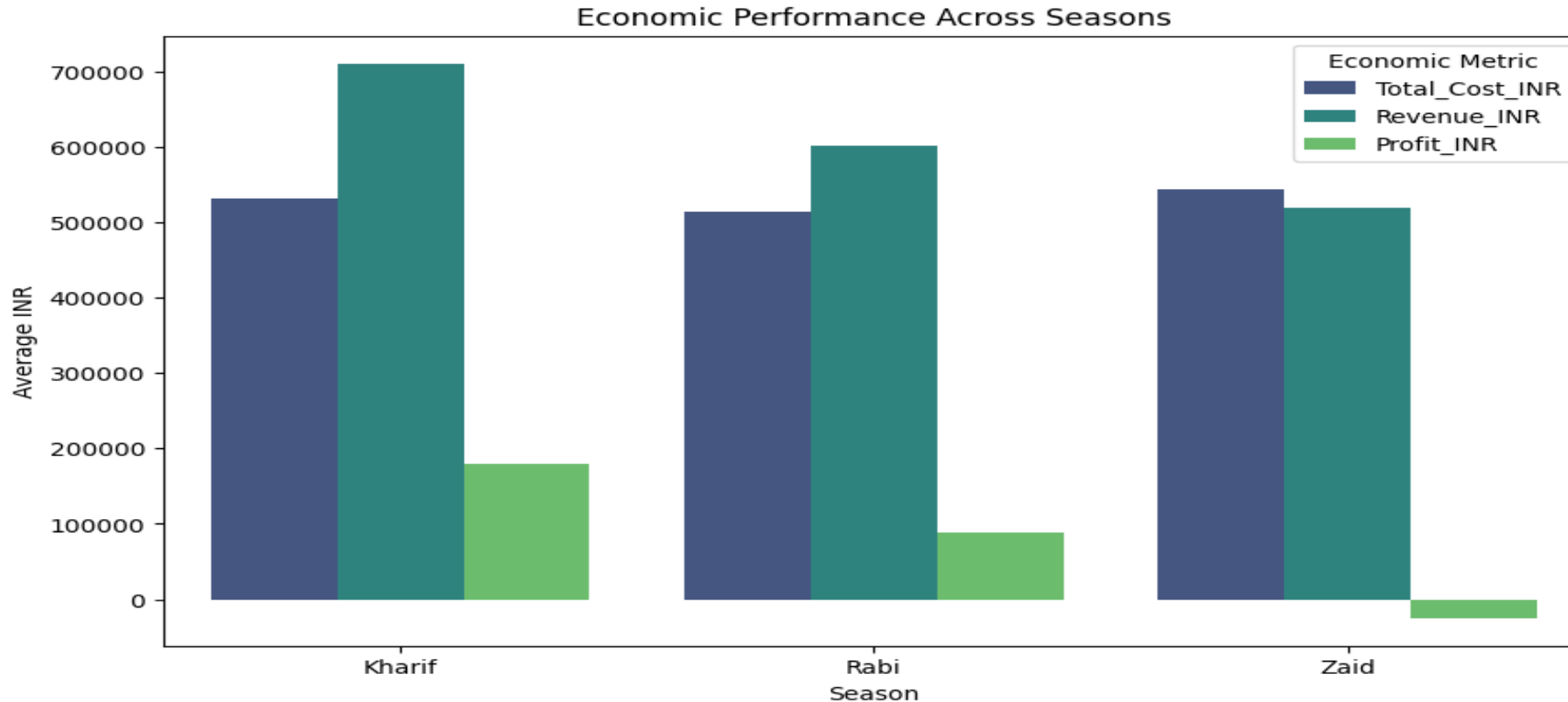
Presentation

- Microsoft PowerPoint

Analytical workflow: Inspect → Clean → Compare → Visualize → Test → Interpret

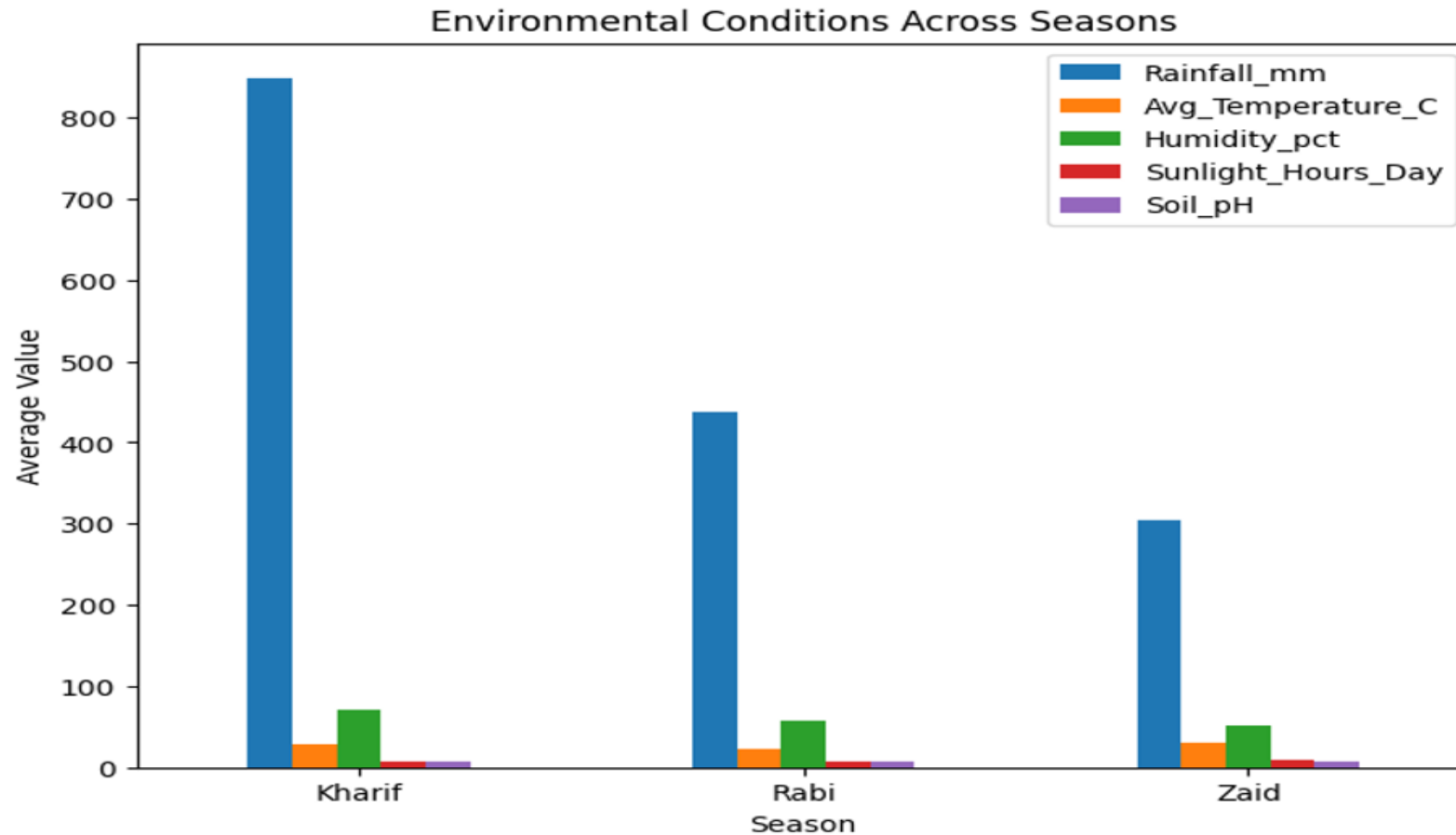


RESULTS



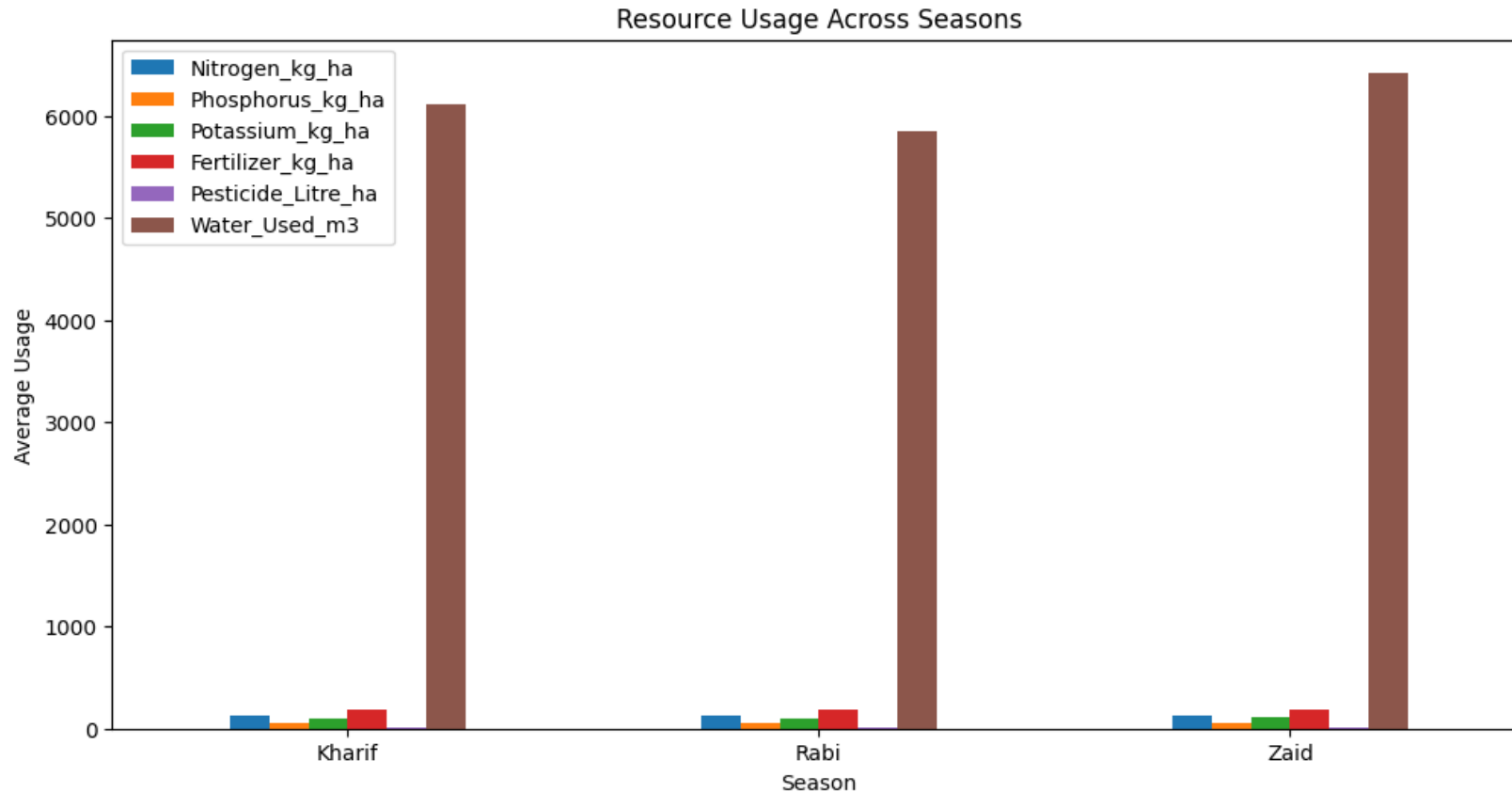
- **Key Insight :** Kharif is the most profitable season, while Zaid operates at a loss due to costs exceeding revenue.

RESULTS



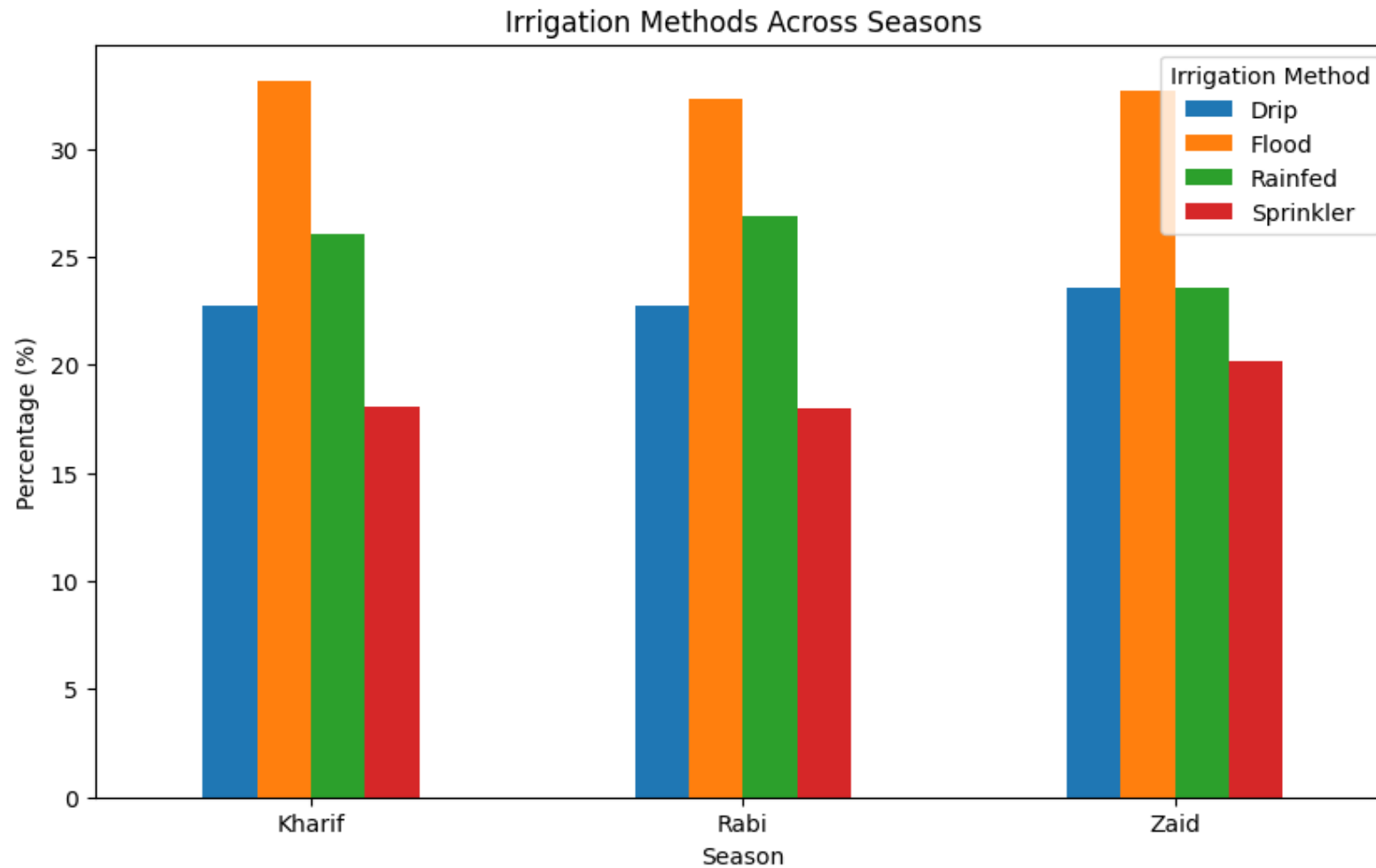
- **Key Insight:** Kharif has the highest rainfall and humidity, while Zaid is the hottest and driest season, highlighting clear seasonal differences in environmental conditions.

RESULTS



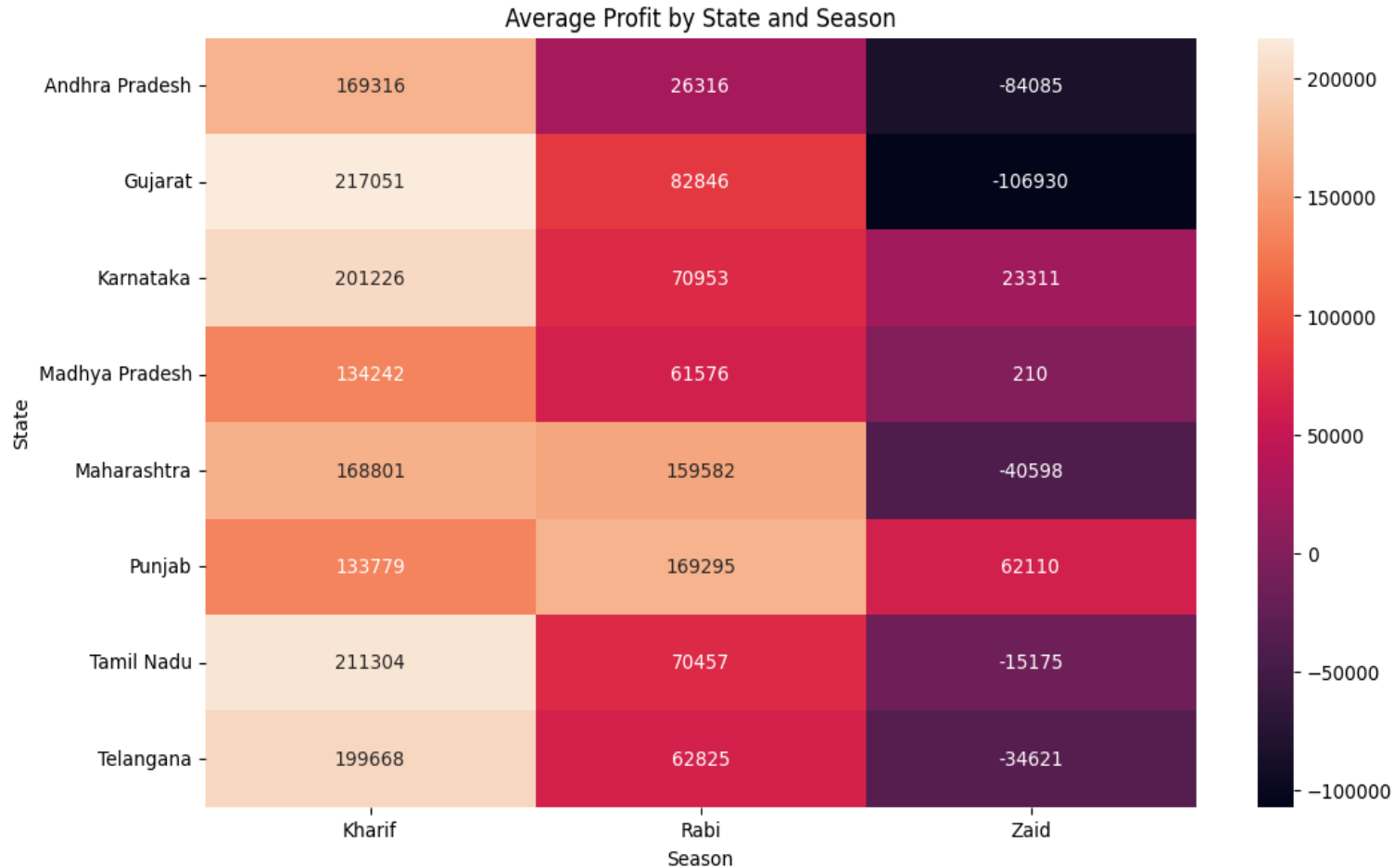
- **Key Insight:** Kharif uses the highest water resources, while Zaid consumes the most water overall, highlighting the need for improved seasonal water management.

RESULTS



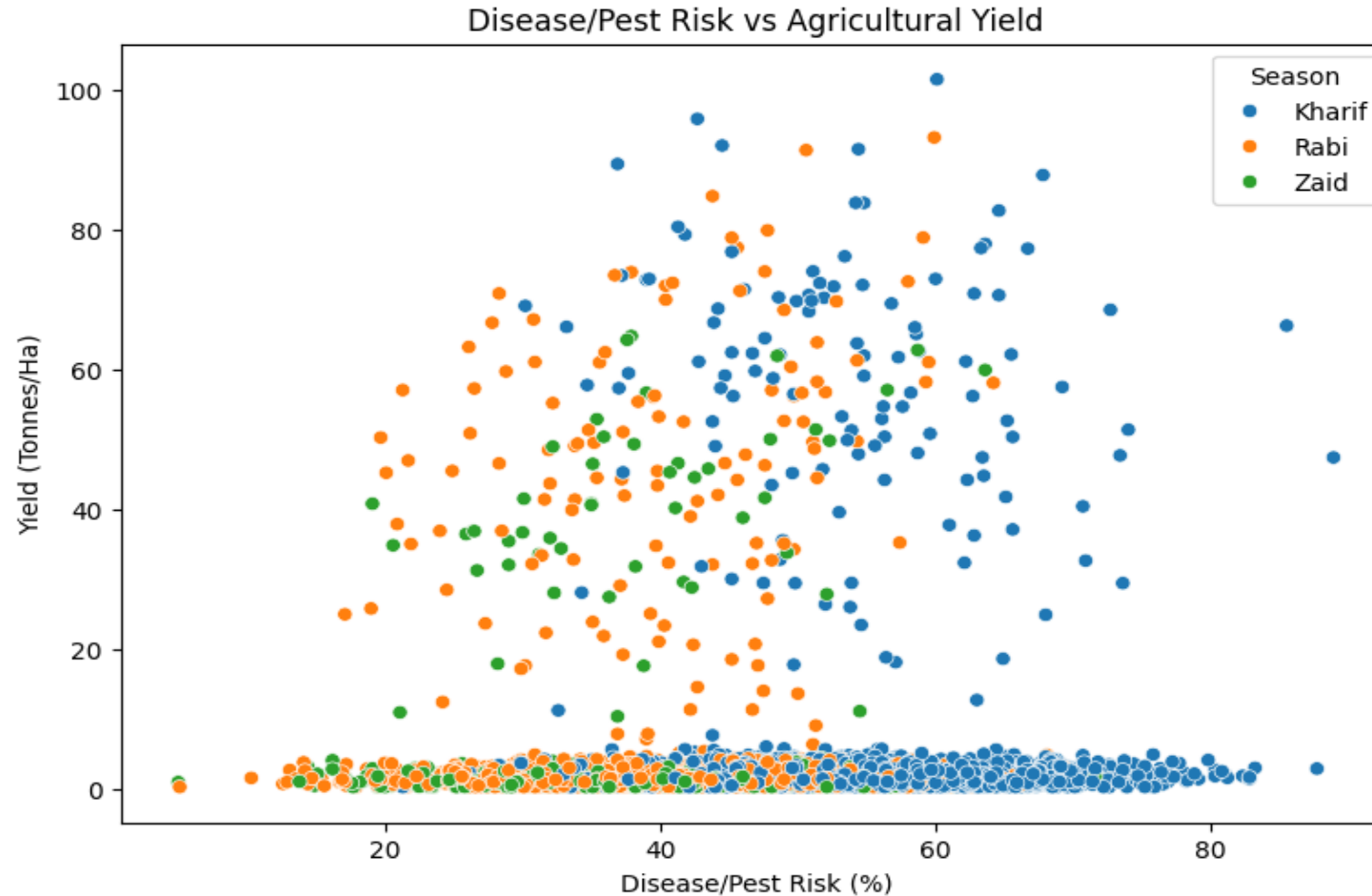
- **Key Insight:** Flood irrigation is the most widely used method across all seasons, while Drip irrigation has a smaller but consistent share.

RESULTS



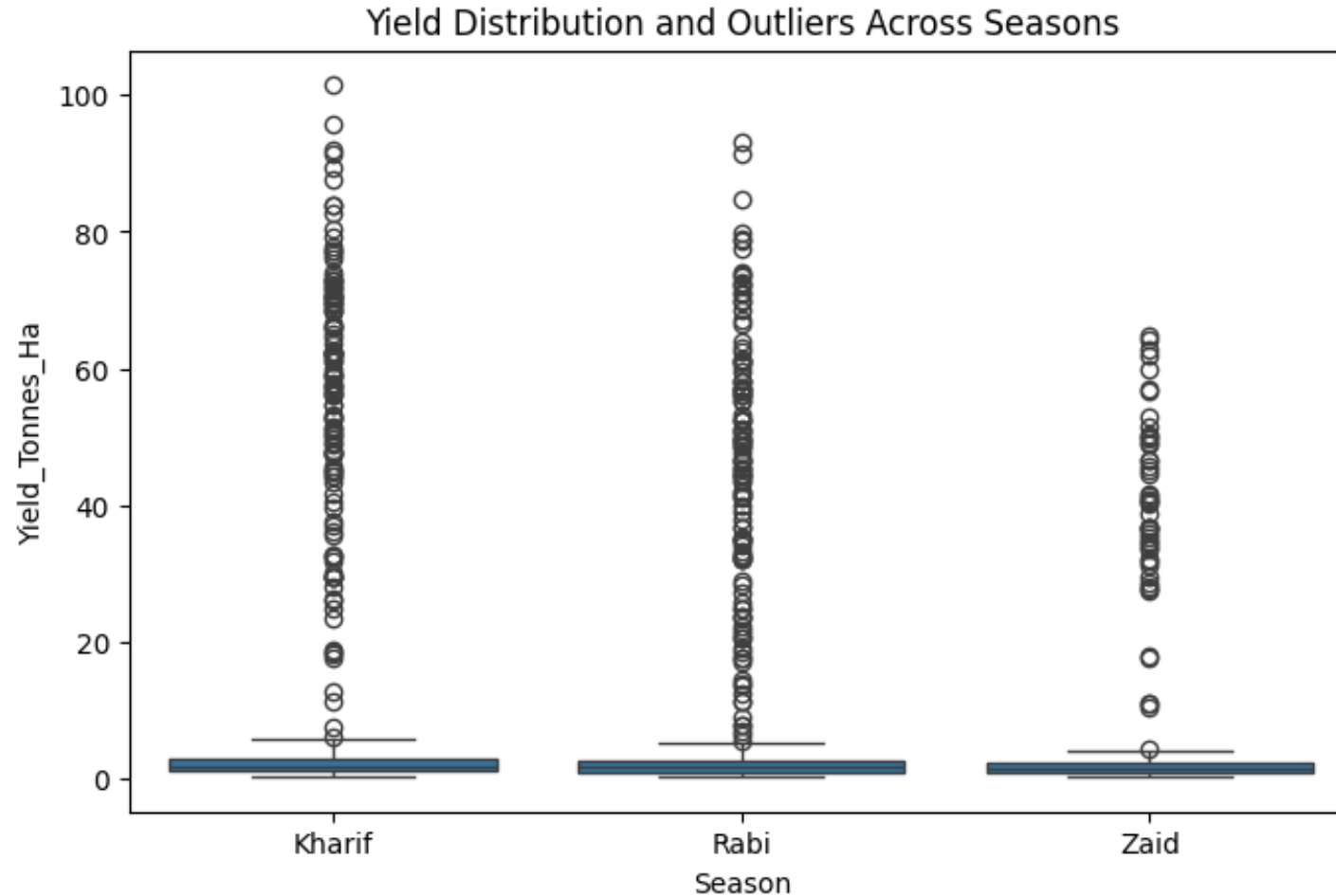
- **Key Insight:** Kharif is the most profitable season across most states, while Zaid records negative profits in several states, showing strong regional and seasonal variation in profitability.

RESULTS



- **Key Insight:** Disease/Pest risk shows variation across seasons, but the relationship with agricultural yield is not consistently strong, indicating that other factors also influence yield.

RESULTS



- **Key Insight:** Kharif shows the widest yield variation and highest outliers, while Zaid has a lower overall yield range, indicating seasonal differences in yield distribution.

Future scope

- **Multi-Year Analysis:** Use data from multiple years to validate seasonal trends and improve reliability. 
- **Predictive Analytics:** Develop machine-learning models to forecast **crop yield, revenue and profitability**.
- **Weather Integration:** Incorporate real-time weather and climate data for better seasonal prediction.
- **Crop & Region Optimization:** Identify the most suitable **crop-season-region combinations** for higher returns.
- **Smart Resource Management:** Optimize **water, fertilizer and irrigation usage** to improve productivity and sustainability.

GitHub Link

<https://github.com/Anirban-Bishnu/Seasonal-Agriculture-Performance-Analysis/tree/main>



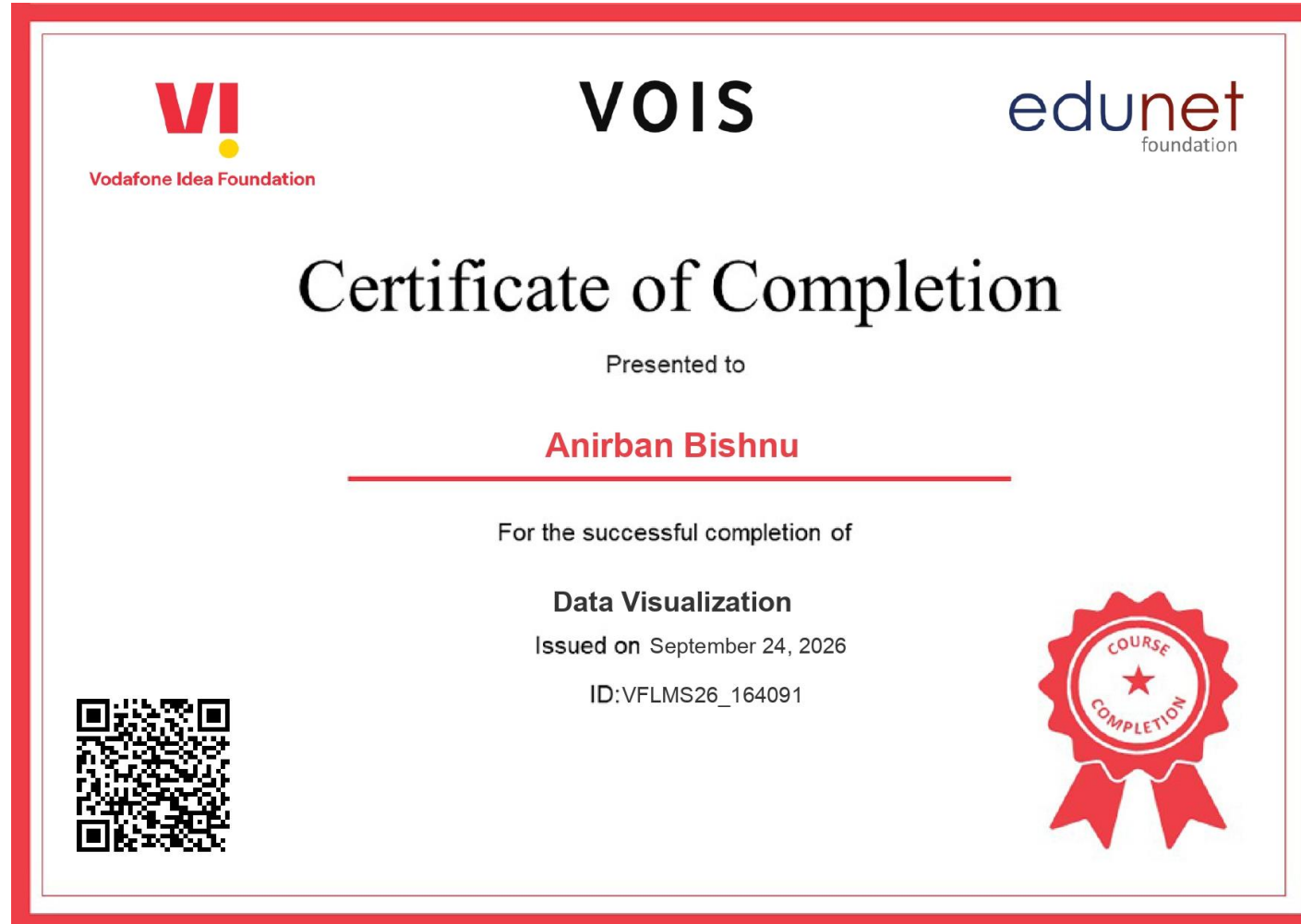
Google Colab Link :

<https://colab.research.google.com/drive/1M4RQRjmjWe2si6QuJ2pzNEyjS4IO3rNT#scrollTo=2nbKqI9rdDVd>



VOIS Course Completion Certificate –

Course name: Data Visualization





Thank you